



Actor Network Theory (ANT)

INFORMATION SYSTEMS THEORIES (IST580S)

DR D LAKAY

Tenets

Actors (or
actant)

Actor-network

Blackbox

Immutable
mobile

Inscription

Irreversibility

Obligatory
passage point
(OPP)

Translation

ANT overview

It is a framework that views society as composed of networks of both **human and non-human actors**.

ANT refers to actors or actants (human and non-human) as **any entity that has the capacity to make a difference**; they have the power to influence outcomes and shape the network.

A network is described by Iyamu (2024:106) as a “group of actors with aligned interests”.

ANT prescribes that a network can be consciously or unconsciously formed (Callon, 1984).

Actors in a network have relationships; they influence each other in various ways, and they collaborate to achieve the network's goals

ANT comprises various concepts/tenets, but one of its key concepts is **translation** (Bencherki, 2017).

Translation

Translation is the central process by which actor-networks are **formed, transformed, and maintained** (Williams, 2020).

Translation is a core process in ANT that describes how the diverse **interests of actors** (both human and non-human) **are aligned and converted into a common interest**, enabling the successful adoption and embedding of a technology (McBride, 2003).

Establishing technology requires aligning the interests of actors within the network. As the interests of actors within the network are aligned, the network becomes stable and the technology is then firmly established. ANT suggests that the aligning of the interests of actors in the network involves the translation of those interests into a common interest in adopting and using the technology

Actors from the outset have a diverse set of interests, and stability rests crucially on the ability to translate, i.e. re-interpret, represent or appropriate others' interests to one's own.

Translation

For Latour (2005), translation means a relationship that doesn't intimate causality, but rather encourages two mediators in a coexistence.

Translation “refers to all the displacements among other actors whose mediation is essential to the occurrence of any action. Instead of a rigid opposition between context and content, the translation chains refer to the work by which the actors change, displace and translate their several and contradictory interests” (Latour, 2001, p. 356).

Law (1992) explains translation as something which implies transformation and the possibility of similarity; the possibility that something (for example, an actor) can represent something else (for example, a network).

How does Translation work?

Actors in a network often have different goals.

Translation involves how these actors with their individual interests can be fulfilled by participating in the network.

This alignment is achieved through common services, etc.

Translation might lead to any of the other tenets, e.g. inscription

Four stages / processes of translation

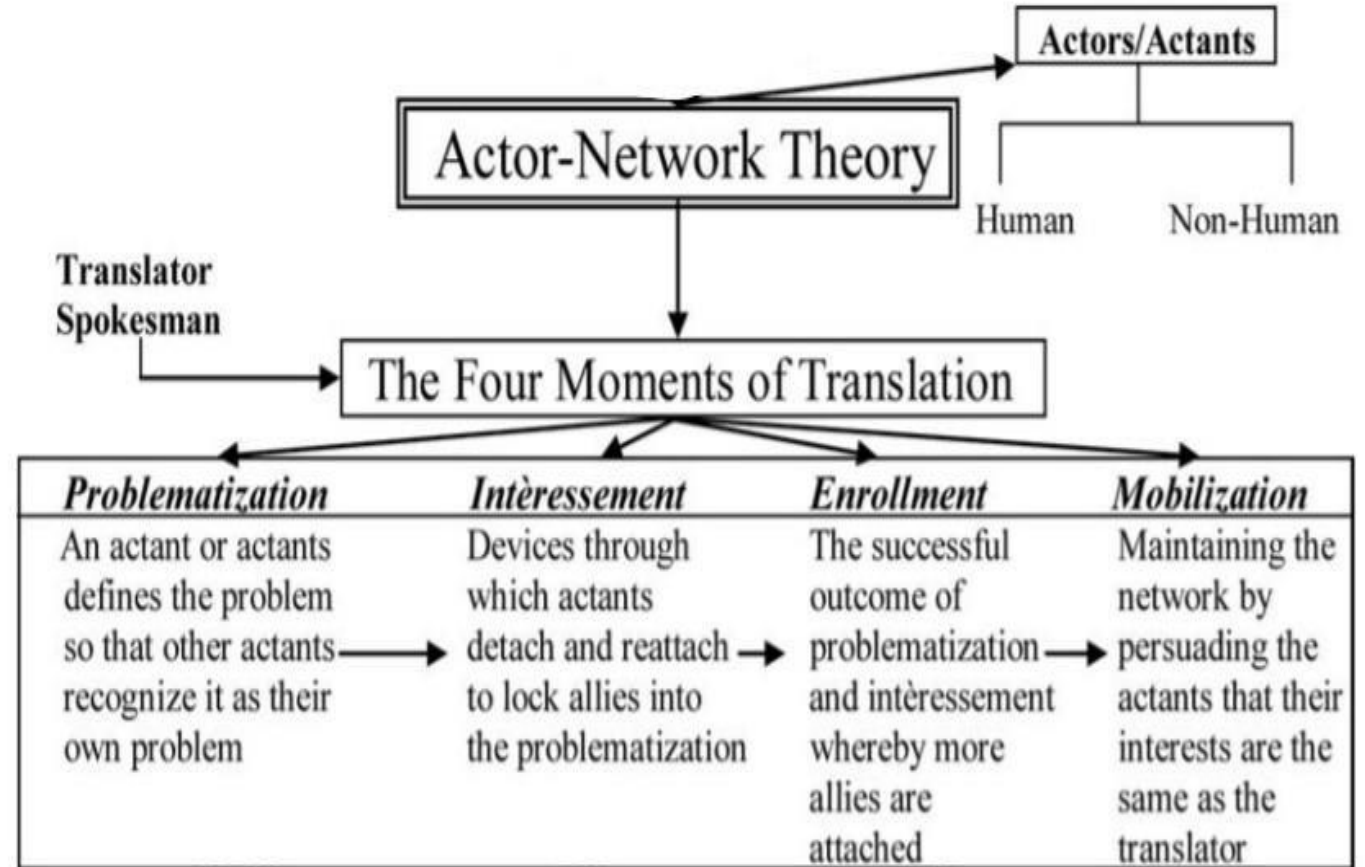
Translation of an actor(s) into a network is achieved through MoT

During the process of problematization, actors define a relevant problem and identify the critical actors.

During interressement, the critical actors try to persuade others to invest in or follow their program.

During enrollment, the critical actors bestow qualities and motivations to actors and establish roles

During mobilization, the formed network gains wider acceptance by making durable and potentially irreversible translations.



Strengths of ANT

3 Principles of ANT

Agnosticism – this means analytical impartiality towards all actors, whether human or non-human, within the phenomena under examination of consideration (Callon, 1986)

Free association – this requires the elimination and abandonment of all a priori distinctions between the technological or natural, and the social aspects (Callon, 1986)

Generalised symmetry – this offers to explain the conflicting viewpoints of different actors in the same terms by using an abstract and neutral vocabulary that works the same way for human and non-human actors. According to Tatnall and Gilding (1999), neither the social nor the technical elements in 'heterogenous networks' should then be given any special explanatory status.

ANT's strengths based on the principles (Iyamu, 2021,79)

Iterative process – it's the process of iterative-ness of actors' interactions and actions over a period and within an environment.

Stability – ANT attempts to answer how a diverse group of actors reaches an agreement at all times; that is, how a social order establishes a certain degree of stability or exhibits structural properties.

Negotiation scheme – ANT provides a fresh perspective on the importance of relationships between actors, which is acquired through negotiation over time.

Process-oriented – the relational and process-oriented sociology of ANT provides a solid theoretical underpinning for unravelling and understanding the human and non-human elements that have on each other (Tatnell & Gilding, 1999).

Power relationship – the offering of a precise and non-functionalist account of how actors become established as powerful entities through the stability of the networks that pass through them.

Systematism – ANT systematically avoids methodological dualism.

Limitations of ANT

Limitations according to Walsham(1997)

Social structures – ANT concentrates on how things get done to the detriment of how broader social constructs shape socio-material practices.

Moral and political issues – ANT destroys the credibility of science by leaving no place for the objective truth that science (allegedly) uncovers. ANT is said to have a disregard for moral and political issues.

Symmetric treatment of humans and non-humans – It uses the same language to describe both the human and non-human actors.

Its tendency to describe as opposed to explain – ANT fails to provide explanations for the dynamic restructuring of networks. ANT also fails to take into account the effects that technology can have on those who are not part of the network and it also fails to support those who are not part of the network.

How to apply ANT

An IT system is an aligned actor-network => The constitutional elements of an IT in a company -> the collection of standards and protocols, uses of the expectations and experiences

Examines the motivations and actions of actors (human and non-human) who form elements, linked by associations of heterogeneous networks of aligned interest.

A major focus of the theory is to try to trace and explain the processes, whereby relatively stable networks of aligned interest are created and maintained or alternatively to examine what such networks fail to establish themselves

1. Objectives – Purpose of the study
2. Identify all the actors – human and non-human
3. Understanding who is aligned => Map relationships of actants and identify relationships.
4. Identify a focal actor and how this focal actor mobilises the other actors/actants.

ANT as a lens

Problematisation

- A focal actor defines (problematizes) a specific problem to other actors, for a possible solution

Intercessionment

- accepts the roles that define them and proposes and lock actors into the problematised programme.

Enrolment

- Actors are assigned tasks in order to propose or find solutions for the problematised initiative.

Mobilisation

- Representatives of all other members of the network, appointed or self appointed.

ANT as a lens - continued

Problematisation

- What problem is defined (problematised)?
- Why was the problem problematised?
- How was the problem problematised?

Intercessment

- Who is interested? Identifies the actors.
- How are the actors interested?
- Why are the actors interested?

Enrolment

- What tasks are assigned to individuals or groups?
- Who assigns the tasks and on what basis?
- How were the tasks assigned?

Mobilisation

- Who are the spokesperson? Identifies the actors.
- How are they appointed?
- Why were the spokesperson appointed?

Why does ANT work (Mcbride, 2003)

ANT has extended across disciplines

This theory has been expanded into a language that supports the joining together of technical and social explanations and description of the phenomenon of human-technology interaction. Provides an explanation of how technology is accepted within communities and organisations

ANT is seen as a way of telling stories about networks of actors - both human and non-human - and about the processes by which technology is established. **Success depends on both technical excellence and social acceptance**

Focuses on heterogeneous networks of actors (aligned interests)

Emphasises translation and inscription (attaching meaning to the technology)

Also describes enrolment and irreversibility

Explains stability and black boxes.

Motto for ANT

FOLLOW THE ACTORS

Thank You for Listening

Any Questions?



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